

The same-row part dies: structural proof for $\lambda = (3, 3, 1^q)$, $q \geq 2$

Clio

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Abstract

The r=1 inductive-reduction paper of 13 May (2026-05-13-r1-inductive-reduction.tex) isolated a clean conjecture (the “same-row-dies conjecture”) whose verification upgrades the r=1 reduction formula to an unconditional statement on every test case. The conjecture was verified computationally across $\lambda = (3, 3, 1^q)$ for $q \in \{2, 3, 4, 5\}$ but lacked a structural proof. We supply that proof here for the family $\lambda = (3, 3, 1^q)$, $q \geq 2$.

Theorem. For $\lambda = (3, 3, 1^q)$ with $q \geq 2$, $n = q + 6$, $\ell = q + 2$, and S the same-row part of $E_{n-1}^+|_{V_\lambda}$ (the span of seminormal vectors v_T for SYTs of λ with letters $n-1, n$ at the same-row pair $(2, 2), (2, 3)$),

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} S = 0.$$

Corollary. The conditional reduction formula (Theorem 9 of the r=1 paper) for $\lambda = (3, 3, 1^q)$ at $q \geq 2$ is now unconditional:

$$B_{\ell+1}^{(\lambda)} V_\lambda = (T_\ell+1)(T_{\ell+1}+1)(T_{n-2}+1) \Phi_* \left(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} \right).$$

Combined with the May-12 evening result on $\mu_1 = (3, 2, 1^{q-1})$ (rigorous at $q \geq 4$, computational at $q = 2, 3$), this gives $\dim B^{(\lambda)} = 2$ structurally at $q \geq 4$.

The proof uses three ingredients: (i) the $H_q(S_{n-2})$ -iso $S \cong V_{(3, 1^{q+1})}$ from Lemma 3.3 of the r=1 paper; (ii) a clean operator commutation identity inside $H_q(S_n)$ via index-gap ≥ 2 commutations; (iii) the May-12 hook j-formula applied to the sub-shape $\nu = (3, 1^{q+1})$ at sharp index. The non-obvious step is (ii): combining R'_{n-2} and R'_{n-1} rearranges into $(T_{n-3}+1)(T_{n-2}+1)R'_{n-3}R'_{n-2}$, exposing two applications of R' -on- S that match the sharp iteration on ν .

1 Setup and notation

Notation follows the r=1 paper (2026-05-13-r1-inductive-reduction.tex): $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$; V_λ is the seminormal Specht module; $E_j^\pm := E_j^\pm(V_\lambda)$ are the $\pm$ -eigenspaces of T_j (eigenvalues $q, -1$);

$$R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad B_j^{(\lambda)} V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

Throughout this paper $\lambda = (3, 3, 1^q)$ with $q \geq 2$, $n = q + 6$, $\ell = q + 2$. The corners of λ are $c_1 = (2, 3)$ (length-preserving) and $c_* = (q+2, 1)$ (column-1 bottom); $\mu_1 := \lambda \setminus \{c_1, c_*\} = (3, 2, 1^{q-1})$.

The **same-row pair** of λ is the pair of cells $(2, 2), (2, 3)$ at row 2 (this is the unique same-row pair of λ , because $\lambda_2 = 3$ and $\lambda_3 = 1 \leq 1 = \lambda_2 - 2$).

Definition 1 (Same-row part S). $S \subseteq V_\lambda$ is the span of $\{v_T : T \in \text{SYT}(\lambda), T(2, 2) = n - 1, T(2, 3) = n\}$. The remaining letters $\{1, \dots, n - 2\}$ fill the cells of the sub-partition

$$\nu := \lambda \setminus \{(2, 2), (2, 3)\} = (3, 1^{q+1}),$$

on $n - 2 = q + 4$ letters. In particular $\dim S = f^\nu$.

Remark 2. The Lemma 3.3 of the r=1 paper gives an explicit $H_q(S_{n-2})$ -equivariant iso $\Psi : S \xrightarrow{\sim} V_\nu$ (under the subalgebra $\langle T_1, \dots, T_{n-3} \rangle$): Ψ sends v_T to $v_{T|_\nu}$, where $T|_\nu$ is the restriction of T to letters $\{1, \dots, n - 2\}$. Equivariance: for $j \leq n - 3$, both sides act by Hoefsmit's rule on positions of $j, j + 1$, which lie in ν (since $j + 1 \leq n - 2$); hence the actions match.

2 The key commutation identity

Lemma 3 (Commutation identity). *Inside $H_q(S_n)$,*

$$R'_{n-2} R'_{n-1} = (T_{n-3} + 1) (T_{n-2} + 1) R'_{n-3} R'_{n-2}. \quad (1)$$

Proof. First, $R'_{n-1} = (T_{n-2} + 1) R'_{n-2}$ by definition. Hence

$$R'_{n-2} R'_{n-1} = R'_{n-2} (T_{n-2} + 1) R'_{n-2}.$$

Next, $R'_{n-2} = (T_{n-3} + 1) R'_{n-3}$, and R'_{n-3} is a product of factors $(T_j + 1)$ with $j \leq n - 4$. Each such T_j has $|j - (n - 2)| \geq 2$, so T_j commutes with T_{n-2} ; equivalently R'_{n-3} commutes with $(T_{n-2} + 1)$. Therefore

$$R'_{n-2} (T_{n-2} + 1) = (T_{n-3} + 1) R'_{n-3} (T_{n-2} + 1) = (T_{n-3} + 1) (T_{n-2} + 1) R'_{n-3}.$$

Substituting:

$$R'_{n-2} R'_{n-1} = (T_{n-3} + 1) (T_{n-2} + 1) R'_{n-3} R'_{n-2}. \quad \square$$

Remark 4. The identity (1) is shape-independent — it holds inside $H_q(S_n)$ as an operator equation, independent of the chosen representation.

3 The hook j-formula on $\nu = (3, 1^{q+1})$

Lemma 5 (May-12 hook j-formula). *For $\nu = (3, 1^{q+1})$ on $m = q + 4$ letters with $q \geq 2$ (so $m \geq 6$),*

$$B_{m-1}^{(\nu)} V_\nu := R_{m-1}'^{(\nu)} R_m'^{(\nu)} V_\nu \subseteq E_1^-|_{V_\nu}. \quad (2)$$

Proof. This is the j-formula for the hook $(3, 1^{m-3})$, proved structurally in 2026-05-12-hook-j-formula.tex (commit 9b532b6). The proof tracks the image of $R_{n-1}' R_n'$ step by step in the seminormal basis: Phase A (R_m') gives $\dim m - 2 = q + 2$ with a basis of T_{m-1} -2-block $+q$ -eigenvectors plus the same-row $+q$ -eigenvector at row 1; Phase B (R_{m-1}') starts with $(T_1 + 1)$ collapsing to $\dim 1$, then a shifting-window argument shows the surviving support is a single SYT v_{T^*} with $T^*(1, 1) = 1$ on letters in $\{n - 2k + 3, \dots, n\}$ (top range), giving the support in E_1^- via min element $\geq 3 \geq 2$. The full Phase B argument lands $B_{m-1}^{(\nu)} V_\nu$ inside E_1^- . $\square$

Remark 6. The full rank-zero theorem $\Pi^{S_m}|_{V_\nu} = 0$ at every $m \geq 6$ follows from (2) by one further R'_{m-2} application: R'_{m-2} leads (on the right, applied first) with $(T_1 + 1)$, which kills E_1^- . We use only (2) here, not the full rank-zero theorem.

4 Main theorem

Theorem 7 (Same-row dies for $(3, 3, 1^q)$). *For $\lambda = (3, 3, 1^q)$ with $q \geq 2$, $n = q + 6$, $\ell = q + 2$, and S the same-row part of Definition 1,*

$$R'_{\ell+1} R'_{\ell+2} \cdots R'_{n-1} S = 0. \quad (3)$$

Proof. Since $\ell + 1 = q + 3 = n - 3$, the iteration in (3) is exactly three R' -applications:

$$R'_{\ell+1} R'_{\ell+2} R'_{n-1} = R'_{n-3} R'_{n-2} R'_{n-1}.$$

We show this acts as zero on S , in three steps.

Step 1. *Apply the commutation identity (1).* We have

$$R'_{n-3} R'_{n-2} R'_{n-1} = R'_{n-3} ((T_{n-3}+1)(T_{n-2}+1) R'_{n-3} R'_{n-2}) = R'_{n-3} (T_{n-3}+1)(T_{n-2}+1) R'_{n-3} R'_{n-2}. \quad (4)$$

Step 2. $R'_{n-3} R'_{n-2}|_S$ lies in $E_1^-|_{V_\lambda}$.

Restrict to S and use the iso $\Psi : S \xrightarrow{\sim} V_\nu$ (Lemma 3.3 of the $r=1$ paper), which is T_j -equivariant for $j \leq n-3$. The operators R'_{n-3}, R'_{n-2} are products of factors $(T_j + 1)$ with $j \leq n-3$, so each is in the $\langle T_1, \dots, T_{n-3} \rangle$ -subalgebra and respects Ψ :

$$\Psi(R'_{n-3}|_S \cdot R'_{n-2}|_S \cdot v) = R'^{(\nu)}_{n-3} R'^{(\nu)}_{n-2} \Psi(v) \quad \text{for all } v \in S.$$

Now $R'^{(\nu)}_{n-2} = R'^{(\nu)}_m$ and $R'^{(\nu)}_{n-3} = R'^{(\nu)}_{m-1}$ (since $m = n-2$). So

$$\Psi(R'_{n-3}|_S \cdot R'_{n-2}|_S \cdot V_\nu) = R'^{(\nu)}_{m-1} R'^{(\nu)}_m V_\nu = B^{(\nu)}_{m-1} V_\nu.$$

By the May-12 hook j -formula (Lemma 5), $B^{(\nu)}_{m-1} V_\nu \subseteq E_1^-|_{V_\nu}$.

Pulling back via Ψ^{-1} (which is T_1 -equivariant since $1 \leq n-3$):

$$R'_{n-3} R'_{n-2} S \subseteq \Psi^{-1}(E_1^-|_{V_\nu}) = E_1^-|_S \subseteq E_1^-|_{V_\lambda}.$$

(The last inclusion uses that T_1 acts on $S \subseteq V_\lambda$ by the same operator whether viewed in S or in V_λ , so $E_1^-|_S = S \cap E_1^-|_{V_\lambda}$.)

Step 3. $(T_{n-3} + 1)(T_{n-2} + 1)$ preserves $E_1^-|_{V_\lambda}$.

For $n \geq 5$, both T_{n-3} and T_{n-2} have index gap ≥ 2 from T_1 , so they commute with T_1 . Hence $(T_{n-3} + 1)$ and $(T_{n-2} + 1)$ commute with T_1 and preserve the T_1 -eigenspaces, in particular $E_1^-|_{V_\lambda}$.

Combining with Step 2,

$$(T_{n-3} + 1)(T_{n-2} + 1) R'_{n-3} R'_{n-2} S \subseteq (T_{n-3} + 1)(T_{n-2} + 1) E_1^-|_{V_\lambda} \subseteq E_1^-|_{V_\lambda}.$$

Step 4. R'_{n-3} kills $E_1^-|_{V_\lambda}$.

$R'_{n-3} = (T_{n-4} + 1)(T_{n-5} + 1) \cdots (T_2 + 1)(T_1 + 1)$. The rightmost factor is $(T_1 + 1)$, which is applied first to any vector. On E_1^- , the operator T_1 acts as -1 , so $(T_1 + 1)|_{E_1^-} = 0$. Hence

$$R'_{n-3} E_1^-|_{V_\lambda} = 0.$$

Conclusion. By (4), Step 3, and Step 4,

$$R'_{n-3} R'_{n-2} R'_{n-1} S = R'_{n-3} ((T_{n-3} + 1)(T_{n-2} + 1) R'_{n-3} R'_{n-2} S) \subseteq R'_{n-3} E_1^-|_{V_\lambda} = 0. \quad \square$$

5 Consequences

Corollary 8 (Reduction formula unconditional for $(3, 3, 1^q)$, $q \geq 2$). *For $\lambda = (3, 3, 1^q)$ with $q \geq 2$,*

$$B_{\ell+1}^{(\lambda)} V_\lambda = (T_{\ell+1} + 1)(T_{\ell+2} + 1)(T_{\ell+3} + 1) \Phi_* \left(B_{\ell(\mu_1)+1}^{(\mu_1)} V_{\mu_1} \right). \quad (5)$$

Proof. By Lemma 4.1 of the r=1 paper (the Φ_* -piece commutation, unconditional) and the Phase A decomposition $E_{n-1}^+|_{V_\lambda} = \Phi_*(V_{\mu_1}) \oplus S$ (Lemma 3.4 of the r=1 paper), it suffices to verify $R'_{\ell+1} \cdots R'_{n-1} S = 0$. This is Theorem 7. $\square$

Corollary 9 (Structural upper bound $\dim B \leq 2$ for $(3, 3, 1^q)$, $q \geq 4$). *For $\lambda = (3, 3, 1^q)$ with $q \geq 4$, $\dim B_{\ell+1}^{(\lambda)} V_\lambda \leq 2$, structurally.*

Proof. By Corollary 8, $\dim B^{(\lambda)} \leq \dim B^{(\mu_1)}$ (since the outer factors and Φ_* are injective on the inner space). For $q \geq 4$, $|\mu_1| = q + 4 \geq 8$, so the May-12 evening theorem applies: $\dim B^{(\mu_1)} = 2$. Hence $\dim B^{(\lambda)} \leq 2$. $\square$

Remark 10 (Lower bound). The matching lower bound $\dim B^{(\lambda)} \geq 2$ is verified computationally for $q \in \{2, 3, 4, 5\}$ (Theorem 12 of the r=1 paper). A structural lower bound via chain decomposition + outer factor injectivity remains open — the proof sketch in `2026-05-13-dim2-331/proof-sketch.md` attempts this but identifies a position-of- $(n - 2)$ invariant that fails to be preserved by the outer factor $(T_{n-2} + 1)$. The structural lower bound is not addressed in this paper.

6 Computational verification

Mod- P verification at $P = 100\,003$, $q = 11$ (the seed specialisation), for $\lambda = (3, 3, 1^q)$ at $q \in \{2, 3, 4\}$:

- Step A (operator identity (1)): **True** at $n \in \{8, 9, 10\}$.
- Step 2 ($R'_{n-3} R'_{n-2} S \subseteq E_1^-$): rank of $(T_1 + 1) \cdot R'_{n-3} R'_{n-2} S = 0$ at $n \in \{8, 9, 10\}$ (verified directly).
- Step 3 ($(T_{n-3} + 1)(T_{n-2} + 1)$ preserves E_1^-): rank of $(T_1 + 1)$ applied to $(T_{n-3} + 1)(T_{n-2} + 1) R'_{n-3} R'_{n-2} S$ is also 0 at $n \in \{8, 9, 10\}$.
- Final image $R'_{\ell+1} R'_{\ell+2} R'_{n-1} S = 0$: direct rank computation gives $\dim 0$ at $n \in \{8, 9, 10, 11\}$.

Scripts: `~/projects/scratch/2026-05-13-same-row-dies/`.

The decay pattern (Phase A) on S matches the rank-zero iteration on $\nu = (3, 1^{q+1})$:

$$\dim S = f^\nu \xrightarrow{R'_{n-1}} q + 2 \xrightarrow{R'_{n-2}} 1 \xrightarrow{R'_{n-3}} 0,$$

with the same dimension drops as $V_\nu \xrightarrow{R'_m} q + 2 \xrightarrow{R'_{m-1}} 1 \xrightarrow{R'_{m-2}} 0$ for ν on $m = q + 4$ letters — exactly the parallel predicted by the iso $S \cong V_\nu + \text{Phase A on } \nu$.

7 Where the proof generalises

The proof of Theorem 7 uses three ingredients, all shape-sensitive:

1. **The $H_q(S_{n-2})$ -iso $S \cong V_\nu$** , where ν is the sub-shape obtained by removing the same-row pair. This is a general fact: it holds whenever λ has a same-row pair (Lemma 3.3 of the r=1 paper).
2. **The commutation identity (1)**. This is purely operator-theoretic in $H_q(S_n)$ and holds independent of the representation, so it transfers to any λ .
3. **The j-formula for ν at sharp index**: $B_{j_0(\nu)}^{(\nu)} V_\nu \subseteq E_1^-|_{V_\nu}$. This is the shape-specific input. For $\nu = (3, 1^{q+1})$ a hook, it is May-12.

Therefore the proof extends to **any** r=1 shape λ with a same-row pair, provided:

- (i) the sub-shape ν has $\ell(\nu) = \ell(\lambda)$ (so that the sharp index $j_0(\nu)$ matches the next R' -application in λ),
- (ii) the j-formula $B_{j_0(\nu)}^{(\nu)} V_\nu \subseteq E_1^-|_{V_\nu}$ is known.

Concrete other cases the proof handles directly.

- Hooks $(k, 1^{n-k})$ with $k \geq 3$: same-row at row 1, $\nu = (k-2, 1^{n-k})$ is a hook with one shorter top row. Condition (i) holds. The j-formula on ν : for $k = 3$, $\nu = (1, 1^{n-3}) = (1^{n-2})$ is the sign rep, trivially in E_1^- ; for $k \geq 4$, $\nu = (k-2, 1^{n-k})$ is again a hook, and one would need to combine with the May-12 hook proof or extend it inductively. (However: hooks $(k, 1^{n-k})$ already have rank-zero proved unconditionally for $k \leq 3$ in May-12 and structurally for all k in the morning's chain decomposition — so this corollary of our proof is not new for hooks.)
- $(k, k, 1^q)$ at $k = 3$: this is the family of this paper.
- $(k, k, 1^q)$ at $k \geq 4$: $\nu = (k, k-2, 1^q)$ is not a hook. The j-formula on ν is not yet structurally established for all such shapes. Stays open under the current proof.

The cleanest next step in this thread is to prove the j-formula for the family $\nu = (k, k-2, 1^q)$ (and more generally non-hook non-r=2-corner shapes), which would propagate the same-row-dies proof through to $(k, k, 1^*)$ at $k \geq 4$.

8 Honest status

What we proved. Same-row-dies for $\lambda = (3, 3, 1^q)$ at every $q \geq 2$, unconditionally, via the $H_q(S_{n-2})$ -iso $S \cong V_{(3, 1^{q+1})}$, an operator commutation, and the May-12 hook j-formula.

What we did not prove. The same-row-dies conjecture for general r=1 shapes with same-row pairs (e.g., $(k, k, 1^*)$ at $k \geq 4$). The obstruction is the j-formula on the non-hook sub-shape ν , which requires its own structural work.

What this closes. The r=1 paper's Theorem 12 ($\dim B = 2$ for $(3, 3, 1^q)$, $q \in \{2, 3, 4, 5\}$, conditional on same-row-dies) is now unconditional at $q \geq 2$. The dim is structural at $q \geq 4$ via May-12 evening on $\mu_1 = (3, 2, 1^{q-1})$; structural plus computational at $q = 2, 3$ pending a structural dim-2 lower bound for $\mu_1 = (3, 2)$ and $(3, 2, 1)$.